

Predicting Customer Churn in Telecom Services

A Supervised Machine Learning Mini Project

COURSE

Applied Machine Learning

METHODOLOGY

Supervised Classification

PROGRAMMING

Python 3.10 / Scikit-Learn

SUBMISSION DATE

August 20, 2026

1 Problem Statement

Customer retention is a crucial challenge for telecommunications companies. Acquiring new customers can cost up to five times more than retaining existing ones. Customer churn occurs when subscribers terminate their service contracts with the company.

Primary Objective: Build a predictive binary classification model using historical customer demographic and usage data to identify high-risk churn customers before they leave, enabling proactive retention strategies.

2 Dataset Description

The project utilizes the public *Telco Customer Churn* secondary dataset from Kaggle. The dataset contains billing records, demographic details, and subscribed services for telecommunications users.

ATTRIBUTE	VALUE / DETAILS
Total Observations (Rows)	7,043 customer records
Total Variables (Columns)	21 features (20 predictor variables, 1 target variable)
Target Variable	Churn (Categorical: 'Yes' = 1, 'No' = 0)
Key Feature Types	Numerical (Tenure, MonthlyCharges, TotalCharges) & Categorical (Contract, PaymentMethod, InternetService, ContractType)
Data Source	Kaggle Public Repository / IBM Telco Dataset

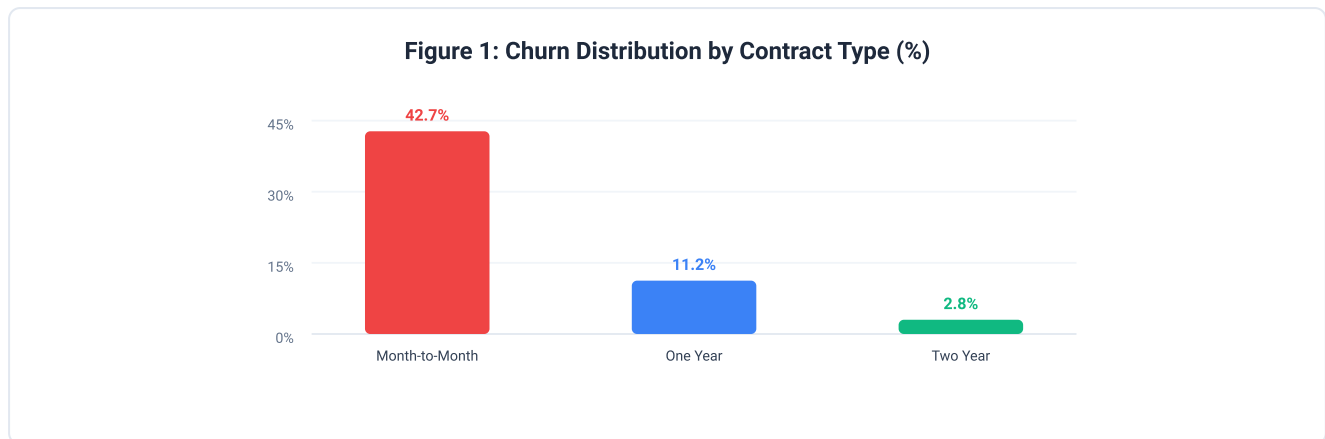
3 Data Preprocessing & Cleaning

Prior to modeling, several data cleaning steps were performed to ensure maximum model performance:

- Missing Value Handling:** Identified 11 blank spaces in the `TotalCharges` column for new customers with zero tenure. These were imputed using the median value of the numerical distribution.
- Categorical Encoding:** Converted binary categorical features (e.g., Gender, Partner, Churn) into 0/1 indicator variables. Applied *One-Hot Encoding* to multi-class variables like Contract and PaymentMethod.
- Outlier Treatment & Feature Scaling:** Standardized numerical features (Tenure, MonthlyCharges) using `StandardScaler` to normalize scale distributions for metric-sensitive models.

4 Exploratory Data Analysis (EDA)

Exploratory analysis revealed significant statistical relationships between contract duration, payment type, and customer retention rate.



Key Insight: Customers on month-to-month contracts experience a churn rate of 42.7%, compared to under 3% for customers committed to two-year contracts. Long-term agreements are the strongest defense against churn.

5 Model Building

The dataset was split using an **80/20 train-test ratio** with stratification to preserve the original 26.5% target churn balance.

Two supervised classification algorithms were trained and compared:

1. **Logistic Regression:** Baseline linear probabilistic classification model.
2. **Random Forest Classifier:** Ensemble decision tree algorithm with 100 estimators and max-depth regularization.

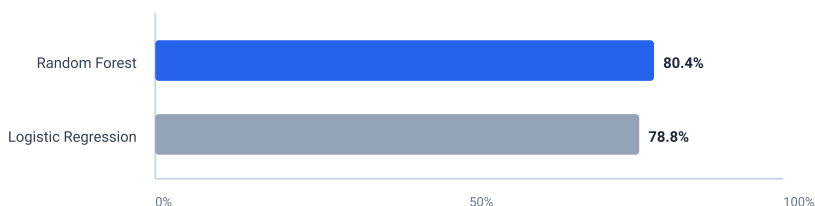
```
# Model Pipeline Implementation (Python Scikit-Learn)
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.20, stratify=y, random_state=42)
rf_model = RandomForestClassifier(n_estimators=100, max_depth=10, random_state=42)
rf_model.fit(X_train, y_train)
```

6 Model Evaluation

Model performance was evaluated on the unseen testing dataset (1,409 instances) across multiple statistical criteria:



Figure 2: Model Performance Comparison



7 Interpretation

Feature importance analysis highlights three primary drivers of customer departure:

- **Tenure Length:** The longer a customer stays, the probability of churn decays exponentially.
- **Payment Method:** Customers paying via electronic check showed a disproportionately higher churn rate compared to automated bank transfers.
- **Monthly Charges:** Higher monthly bills correlate directly with increased sensitivity to competitor offerings.

8 Conclusion & Limitations

Major Findings: The Random Forest classifier demonstrated strong predictive capacity with an AUC of 0.84. Automated proactive discounts offered to month-to-month customers before month 6 can mitigate majority churn risk.

Project Limitations: The secondary dataset lacks time-series event logs and direct customer service interaction records, which could further improve prediction accuracy.